

Math 241, F12
Quiz 4

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (4 points) Compute

$$\begin{aligned} & \lim_{t \rightarrow +\infty} \langle \arctan t, e^{-2t}, \frac{\ln t}{t} \rangle \\ &= \langle \lim_{t \rightarrow +\infty} \arctan t, \lim_{t \rightarrow +\infty} e^{-2t}, \lim_{t \rightarrow +\infty} \frac{\ln t}{t} \rangle \\ &= \langle \frac{\pi}{2}, 0, \lim_{t \rightarrow +\infty} \frac{\frac{1}{t}}{1} \rangle \quad \leftarrow \text{L'Hospital's rule} \\ &= \langle \frac{\pi}{2}, 0, 0 \rangle \end{aligned}$$

Problem 2. (6 points) Let $\mathbf{r}(t) = \sqrt{2}t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}$.

(a) Find the unit tangent vector $\mathbf{T}(t)$ of the curve at the point with $t = 0$.

$$\begin{aligned} \mathbf{r}'(t) &= \sqrt{2}\mathbf{i} + e^t\mathbf{j} - e^{-t}\mathbf{k} \\ \Rightarrow \mathbf{r}'(0) &= \sqrt{2}\mathbf{i} + e^0\mathbf{j} - e^0\mathbf{k} = \sqrt{2}\mathbf{i} + \mathbf{j} - \mathbf{k} \\ |\mathbf{r}'(0)| &= \sqrt{(\sqrt{2})^2 + 1^2 + (-1)^2} = \sqrt{4} = 2 \\ \Rightarrow \mathbf{T}(0) &= \frac{\mathbf{r}'(0)}{|\mathbf{r}'(0)|} = \frac{\langle \sqrt{2}, 1, -1 \rangle}{2} = \langle \frac{\sqrt{2}}{2}, \frac{1}{2}, -\frac{1}{2} \rangle \text{ or } \frac{\sqrt{2}}{2}\mathbf{i} + \frac{1}{2}\mathbf{j} - \frac{1}{2}\mathbf{k} \end{aligned}$$

(b) Find the length of the curve defined by $\mathbf{r}(t)$ with $0 \leq t \leq 1$.

$$\begin{aligned} |\mathbf{r}'(t)| &= \sqrt{(\sqrt{2})^2 + (e^t)^2 + (e^{-t})^2} = \sqrt{2 + (e^t)^2 + (e^{-t})^2} \\ &= \sqrt{(e^t + e^{-t})^2} = e^t + e^{-t} \\ L &= \int_0^1 |\mathbf{r}'(t)| dt = \int_0^1 e^t + e^{-t} dt \\ &= (e^t - e^{-t}) \Big|_{t=0}^{t=1} \\ &= (e^1 - e^{-1}) - (e^0 - e^0) \\ &= e - \frac{1}{e} \end{aligned}$$